PROJECT

ON

SMALL OFFICE NETWORK DESIGN AND CONFIGURATION

USING CISCO PACKET TRACER

Submitted in partial fulfilment of the course requirements

Submitted by

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&

Abhijeet shinde

CERTIFICATE

This is to certify that the project entitled “Small Office Network Design and Configuration Using Cisco Packet Tracer” has been successfully completed by Mohd Faiz Khan as a part of the course requirements of the Anudip guidance & placement programme.

The work presented in this project is based on the design, configuration, implementation and testing of a small office network consisting of separate Engineering, Sales and Office Server network segments.

Project Guide

Signature: _____

Date: _____

Head of Department

Signature: _____

Place: _____

DECLARATION

I, Mohd Faiz Khan, hereby declare that the project titled "Small Office Network Design and Configuration Using Cisco Packet Tracer" is my own academic work. The network design, configuration steps, testing and observations documented in this black book have been prepared for educational purposes.

I further declare that the screenshots and topology illustrations included in this report are taken from the Packet Tracer project files and configuration work supplied with this submission.

Place: _____

Date: _____

Student Signature: _____

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my project guide and the faculty members for their guidance and support throughout the completion of this project. Their suggestions helped me understand practical networking concepts and apply them through Cisco Packet Tracer.

I also thank my institution, Anudip foundation, for providing the academic environment and resources required to complete this work. Finally, I am grateful to everyone who supported me during the planning, configuration and testing stages of this project.

ABSTRACT

This project presents the design and simulation of a small office network for a startup organization using Cisco Packet Tracer. The office has two main departments—Engineering and Sales—and a separate network segment for shared office-server resources. The primary objective is to keep departmental traffic logically separated while allowing authorized communication between departments and access to shared services.

The network is built around a central Layer 2 switch and a router configured for inter-VLAN routing using the router-on-a-stick method. Three VLANs are created: VLAN 10 for Engineering, VLAN 20 for Sales and VLAN 30 for the Office Server. The switch assigns end-device ports to the appropriate VLANs and uses an IEEE 802.1Q trunk to carry VLAN traffic to the router. Router subinterfaces act as default gateways for the three IP networks.

The project demonstrates practical skills in network topology design, VLAN creation, access-port assignment, trunk configuration, IP addressing, router subinterfaces, device connectivity and troubleshooting. Connectivity is verified using interface status information and end-to-end ping tests in the simulation.

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1. INTRODUCTION

A computer network enables devices within an organization to exchange information and share resources. As an office grows, placing every device in one flat network can make administration difficult and may expose resources to unnecessary traffic. Virtual LANs (VLANs) provide logical separation on a switched network without requiring physically separate switches.

This project simulates a small startup office network. Engineering and Sales are placed in different VLANs and IP subnets. A third VLAN is reserved for a shared office server. A router provides communication between these networks through separate subinterfaces connected to a trunk port on the switch.

2. PROBLEM STATEMENT

The organization requires a simple and manageable office network in which Engineering and Sales devices operate on separate subnets. At the same time, users should be able to communicate across departments when required and access a shared office server. The network must be easy to expand, configure and troubleshoot.

3. OBJECTIVES

- Design a small office network using Cisco Packet Tracer.
- Create separate VLANs for Engineering, Sales and the Office Server.
- Assign switch access ports to the correct VLANs.
- Configure an 802.1Q trunk between the switch and router.
- Configure router subinterfaces to provide inter-VLAN routing.
- Assign IP addresses and default gateways to end devices.
- Verify connectivity using ping tests and interface status commands.

4. SCOPE OF THE PROJECT

The project covers a simulated local area network (LAN) for a small office. It includes logical segmentation, IPv4 addressing, switching, router-on-a-stick inter-VLAN routing and basic connectivity validation. The Packet Tracer simulation is intended as a learning model and does not represent a production security assessment or a full enterprise deployment.

5. TOOLS AND TECHNOLOGIES

- Cisco Packet Tracer – network topology creation and simulation.
- Cisco 2960 Layer 2 switch – VLAN and access/trunk port configuration.
- Cisco 2911 router (or equivalent Packet Tracer router) – inter-VLAN routing.
- Ethernet links – connectivity between end devices, switch and router.
- IPv4 addressing and subnet masks – logical addressing of each network.
- ICMP / ping – basic reachability testing.

6. NETWORK REQUIREMENTS AND TOPOLOGY

The simulated office contains Engineering workstations, Sales workstations, a shared Office Server and central network equipment. The central switch provides Layer 2 connectivity. A single physical router-to-switch link carries the three VLANs as tagged traffic. The router uses one logical subinterface per VLAN.

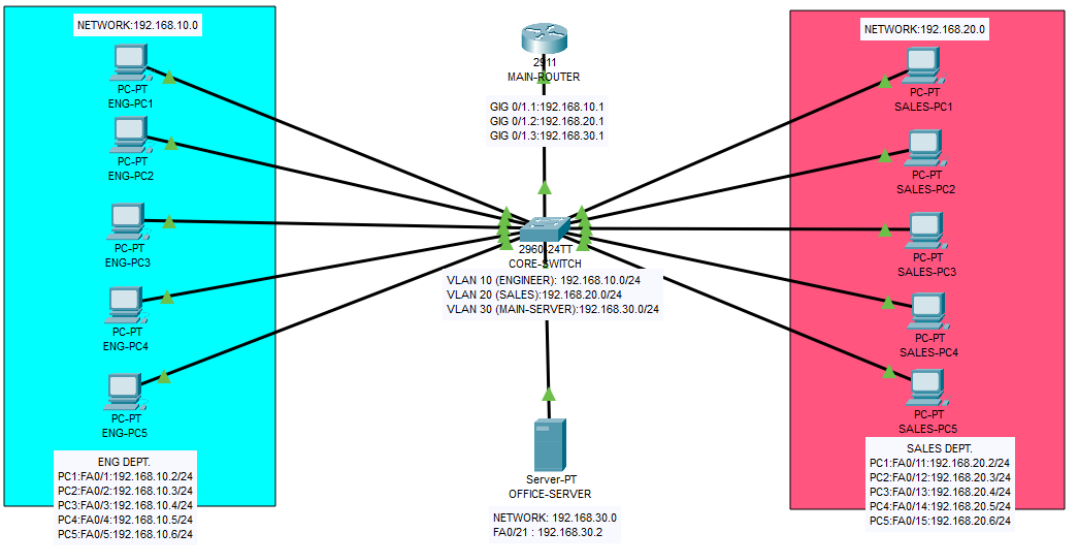


Figure 1: Overall Packet Tracer topology showing department segments, central switching and routing.

Logical VLAN allocation:

VLAN ID	VLAN Name	Purpose
10	ENGINEERING	Engineering department devices
20	SALES	Sales department devices
30	OFFICE-SERVER	Shared office server segment

7. IP ADDRESSING SCHEME

The router interface status screenshot shows the following gateway addresses. Each network uses a /24 subnet mask (255.255.255.0). End-device addresses should be unique within their subnet and use the corresponding router subinterface address as the default gateway.

Department / VLAN	Network	Subnet Mask	Default Gateway	Router Interface
Engineering / 10	192.168.10.0/24	255.255.255.0	192.168.10.1	G0/1.1
Sales / 20	192.168.20.0/24	255.255.255.0	192.168.20.1	G0/1.2
Office Server / 30	192.168.30.0/24	255.255.255.0	192.168.30.1	G0/1.3

Example end-device addressing (illustrative allocation; use the exact addresses configured in the Packet Tracer file):

- Engineering PCs: 192.168.10.11–192.168.10.15; gateway 192.168.10.1.
- Sales PCs: 192.168.20.11–192.168.20.15; gateway 192.168.20.1.
- Office Server: 192.168.30.10; gateway 192.168.30.1.

8. NETWORK IMPLEMENTATION AND CONFIGURATION

8.1 Create VLANs on the switch

The switch is assigned the hostname CORE-SWITCH. VLAN 10, VLAN 20 and VLAN 30 are created and given descriptive names.

```
enable
configure terminal
hostname CORE-SWITCH
vlan 10
 name ENGINEERING
exit
vlan 20
 name SALES
exit
vlan 30
 name OFFICE-SERVER
exit
end
show vlan brief
```

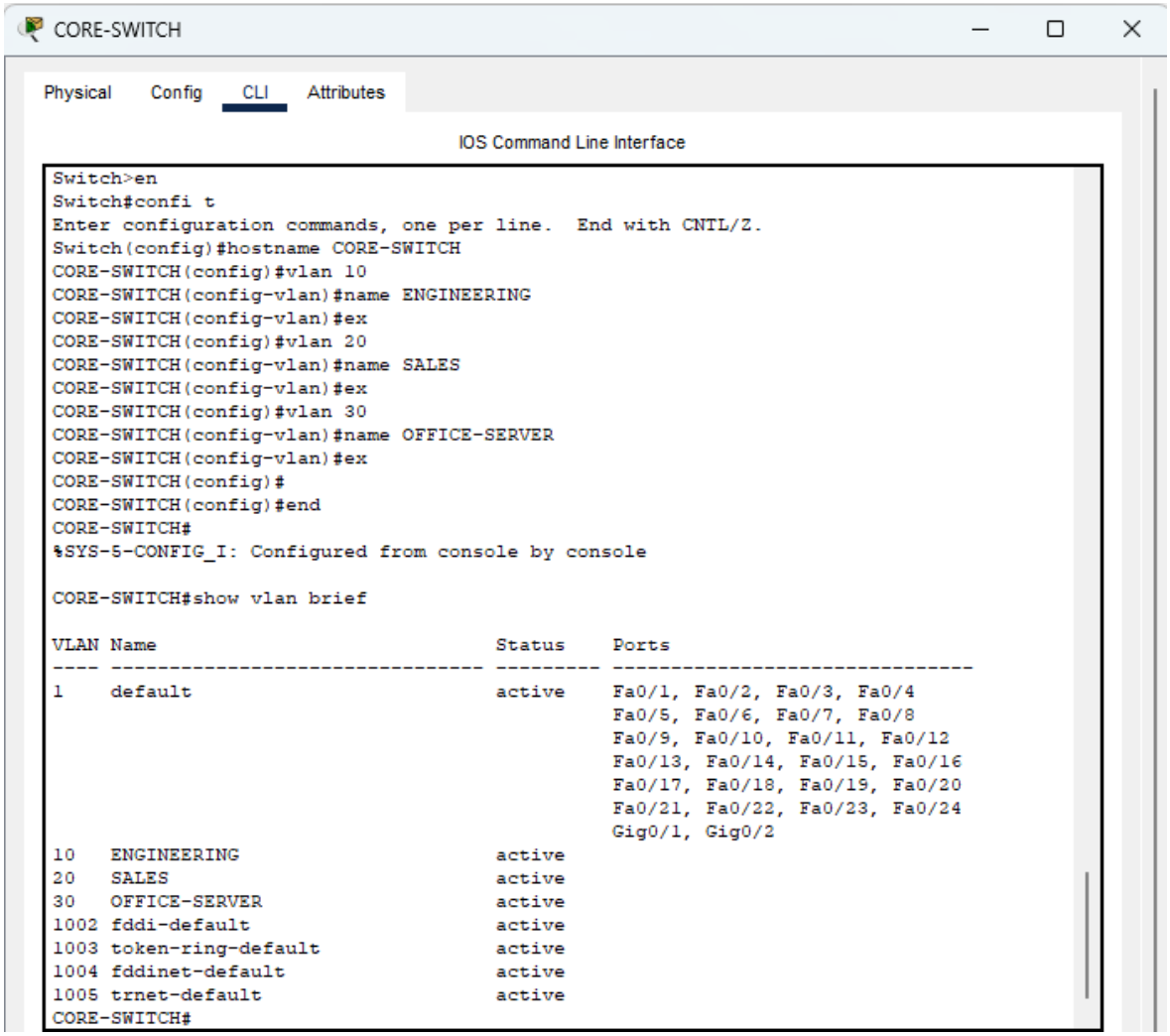


Figure 2: VLAN creation and verification using show vlan brief.

8.2 Assign access ports

The switch access ports are assigned to the VLAN matching the connected department. In the supplied configuration evidence, FastEthernet0/1–0/10 are assigned to Engineering, FastEthernet0/11–0/20 to Sales, and FastEthernet0/21 to the Office Server VLAN.

```
configure terminal
interface range fa0/1-10
 switchport mode access
 switchport access vlan 10
exit
interface range fa0/11-20
 switchport mode access
 switchport access vlan 20
exit
```

```
interface fa0/21
 switchport mode access
 switchport access vlan 30
exit
```

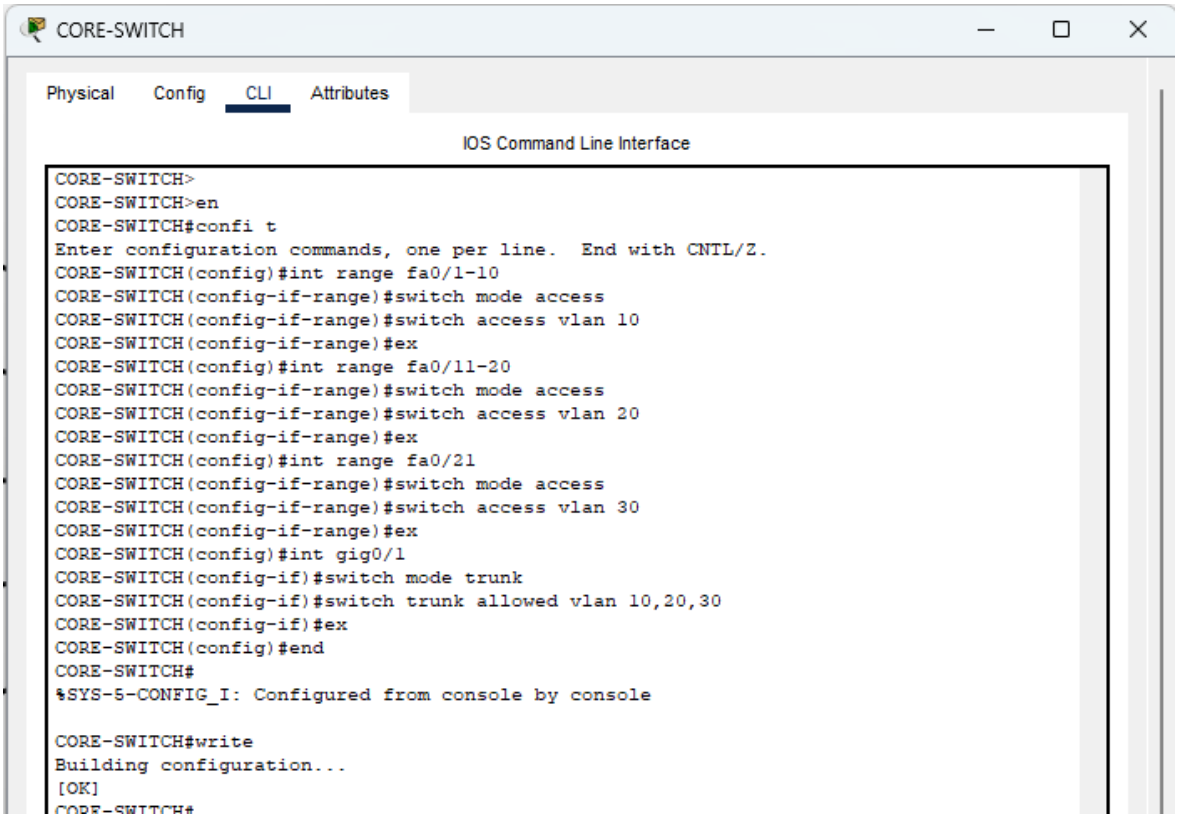


Figure 3: Access-port assignment and trunk configuration on CORE-SWITCH.

8.3 Configure the trunk link

The switch uplink connected to the router is configured as an 802.1Q trunk. The trunk allows VLANs 10, 20 and 30 to pass between the switch and router.

```
interface gigabitEthernet0/1
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30
 no shutdown
end
show interfaces trunk
```

8.4 Enable the router interface

The router physical interface connected to the switch is enabled. The physical interface carries tagged traffic for all three VLANs.

```
enable
configure terminal
hostname MAIN-ROUTER
interface gigabitEthernet0/1
 no shutdown
```

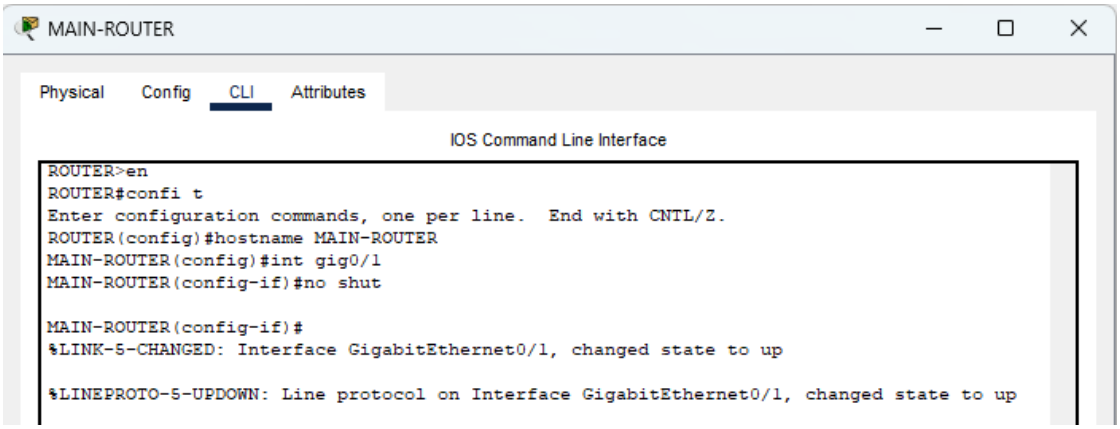


Figure 4: Enabling the router interface connected to the switch.

8.5 Configure router subinterfaces

Three router subinterfaces are created. Each subinterface uses dot1Q encapsulation for its VLAN and is assigned the gateway IP address for that VLAN.

```
interface gigabitEthernet0/1.1
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
exit
interface gigabitEthernet0/1.2
 encapsulation dot1Q 20
 ip address 192.168.20.1 255.255.255.0
exit
interface gigabitEthernet0/1.3
 encapsulation dot1Q 30
 ip address 192.168.30.1 255.255.255.0
exit
end
show ip interface brief
```

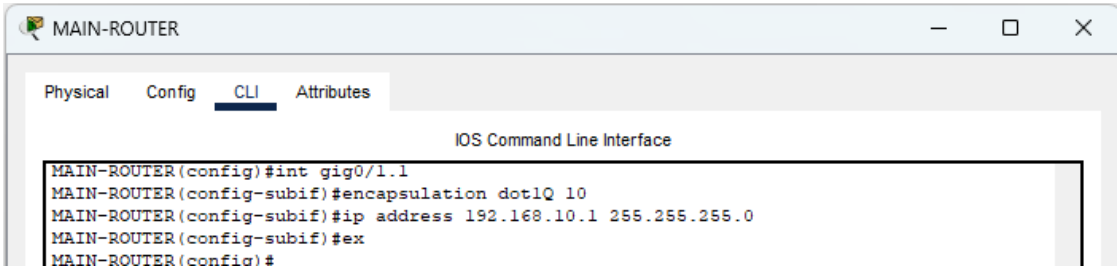


Figure 5: VLAN 10 subinterface configuration.

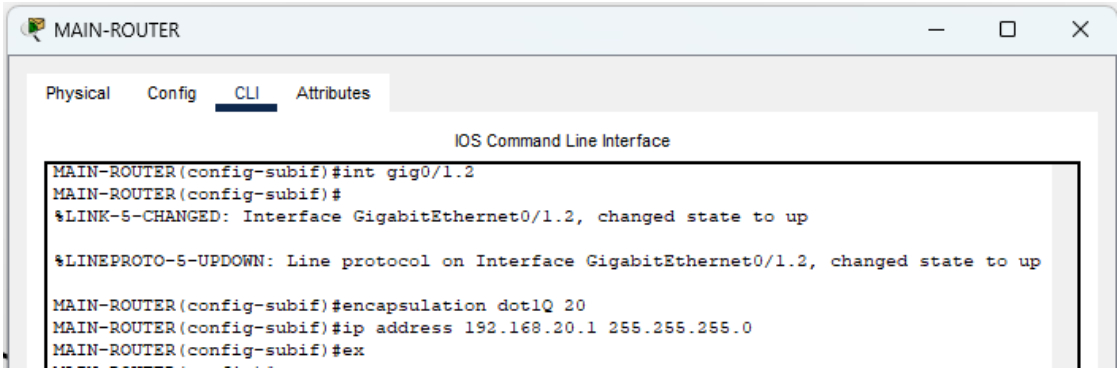


Figure 6: VLAN 20 subinterface configuration.

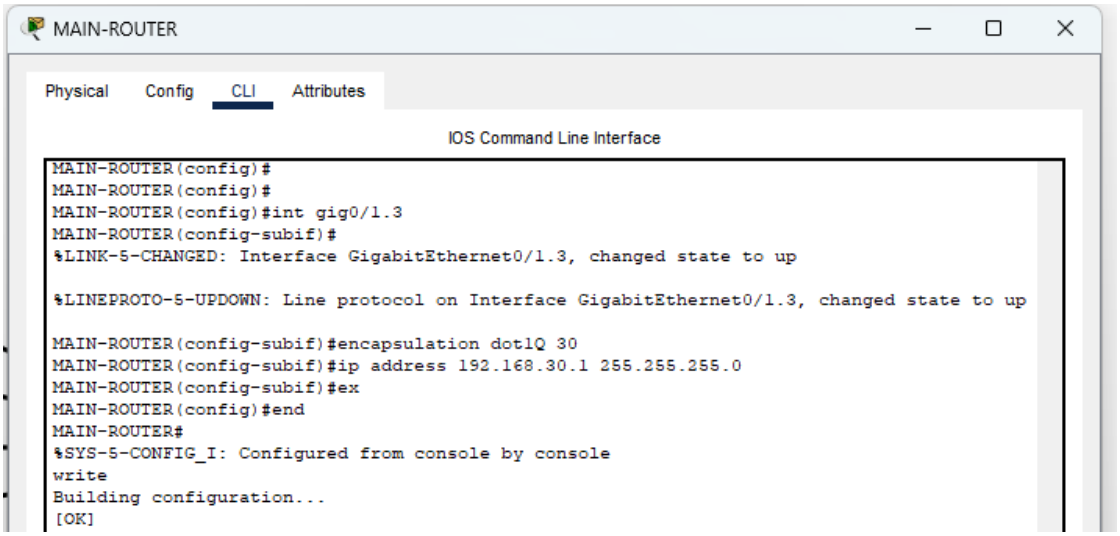


Figure 7: VLAN 30 subinterface configuration.

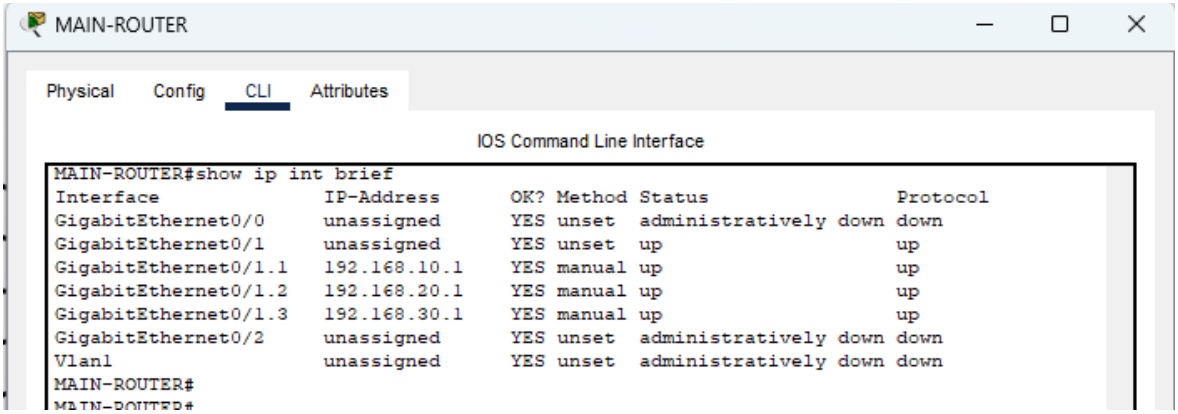


Figure 8: Router subinterfaces shown as up/up with their gateway IP addresses.

8.6 Configure end devices

Each PC is configured with a static IPv4 address, subnet mask and default gateway according to its department VLAN. The server is assigned an address from the Office Server subnet. The gateway value must match the router subinterface for the relevant VLAN.

9. TESTING AND RESULTS

The network is tested in Packet Tracer after the VLAN, trunk, router and end-device configuration is complete. The main checks are:

1. Verify VLAN membership using show vlan brief on the switch.
2. Verify the trunk link and allowed VLAN list using show interfaces trunk.
3. Verify router subinterfaces and IP addresses using show ip interface brief.
4. Ping the default gateway from each PC.
5. Ping between Engineering and Sales devices to validate inter-VLAN routing.
6. Ping the Office Server from devices in both departments to validate shared-resource reachability.

The supplied evidence includes successful interface state indications and command-line testing screenshots. For a final practical demonstration, repeat the ping tests from each department PC and record the actual results in the table below.

Test	Source	Destination	Result
Gateway reachability	Engineering PC	192.168.10.1	To be verified
Gateway reachability	Sales PC	192.168.20.1	To be verified
Inter-VLAN	Engineering PC	Sales PC	To be verified
Server access	Sales / Engineering PC	192.168.30.10	To be verified

10. ADVANTAGES AND LIMITATIONS

Advantages

- Departmental separation improves logical organization and simplifies administration.
- A single trunk link carries multiple VLANs, reducing the number of physical router links required.
- Centralized routing allows communication between VLANs when permitted.
- The design can be expanded by adding more access ports or VLANs.

Limitations

- The design uses a single router and switch, so either device can become a single point of failure.
- The simulation does not include production-grade firewall policies, access control lists or redundancy.
- Static addressing requires manual administration unless DHCP is added.
- Successful reachability does not by itself prove that the network is secure.

11. CONCLUSION AND FUTURE SCOPE

The project successfully demonstrates the design and configuration approach for a small office LAN using Cisco Packet Tracer. Separate VLANs and IPv4 subnets are used for Engineering, Sales and shared server resources. The switch handles VLAN segmentation, while router subinterfaces provide inter-VLAN routing over an 802.1Q trunk.

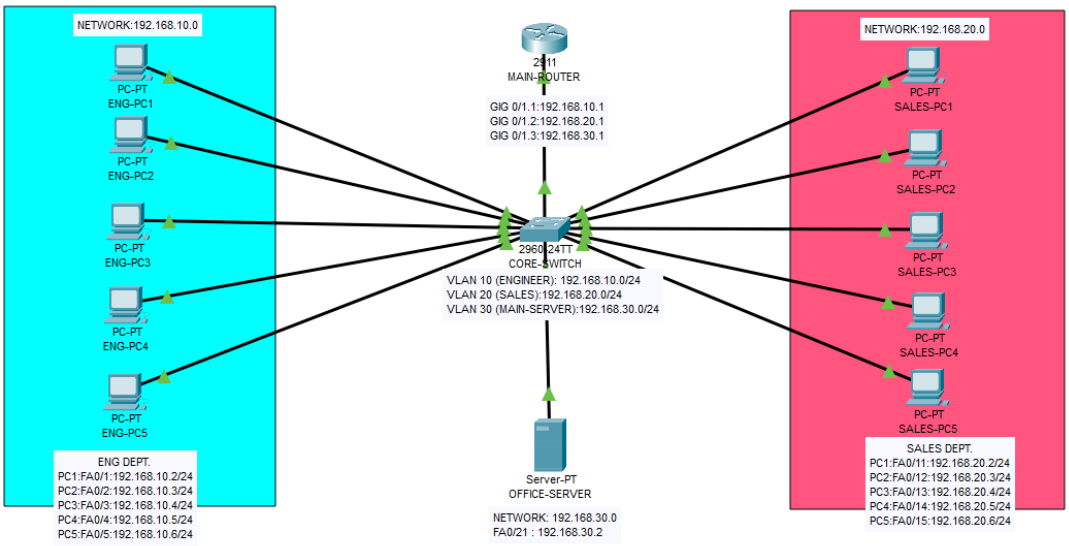
Future improvements may include DHCP services, access control lists to restrict inter-department traffic, firewall deployment, network address translation for internet access, switch and router redundancy, centralized monitoring and documented backup procedures.

12. REFERENCES

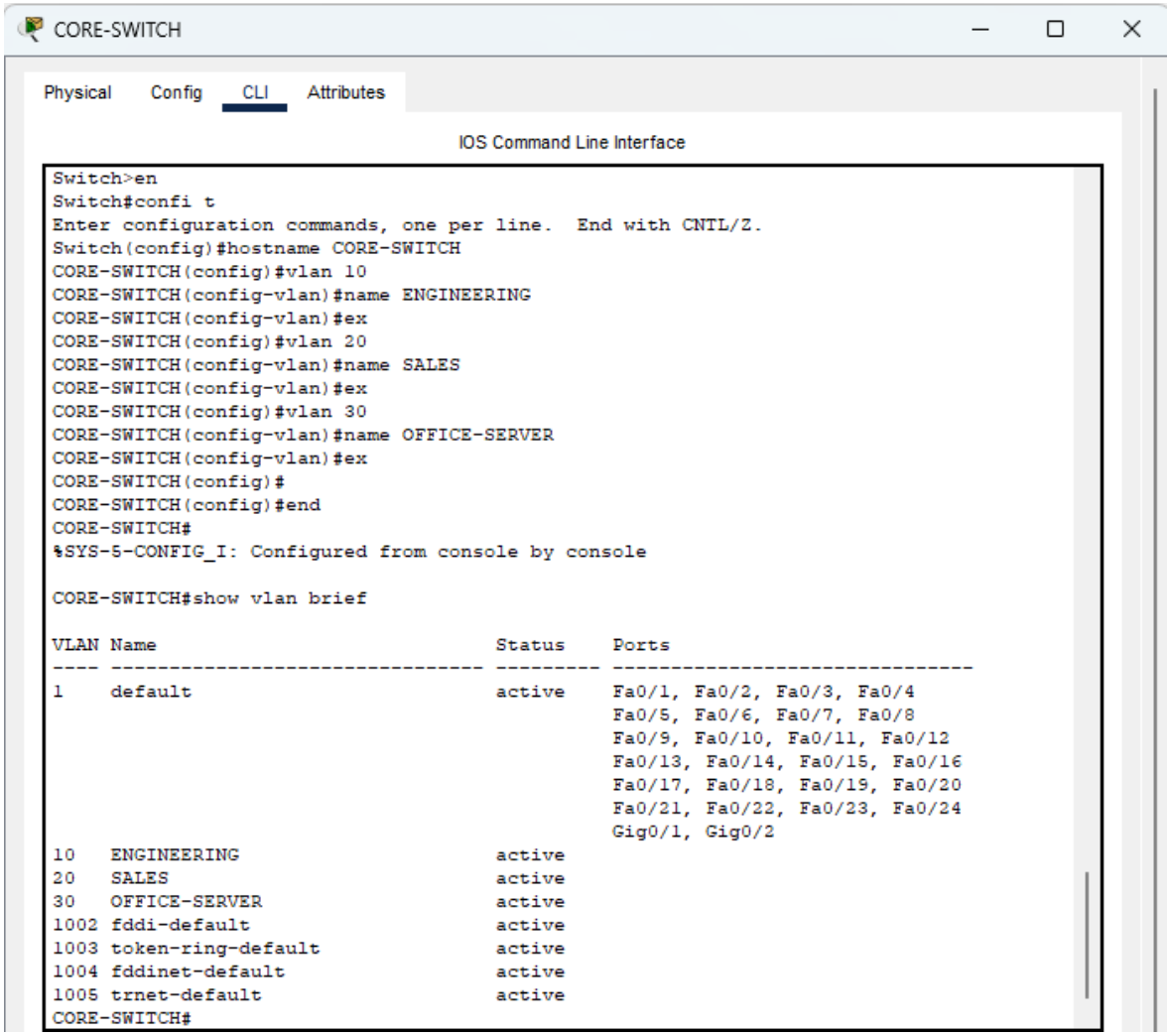
- Cisco Networking Academy – Cisco Packet Tracer learning resources.
- Cisco IOS command reference – VLAN, switchport, trunk and router subinterface configuration.
- Course notes on IPv4 addressing, subnetting, VLANs and inter-VLAN routing.

13. APPENDIX – CONFIGURATION EVIDENCE

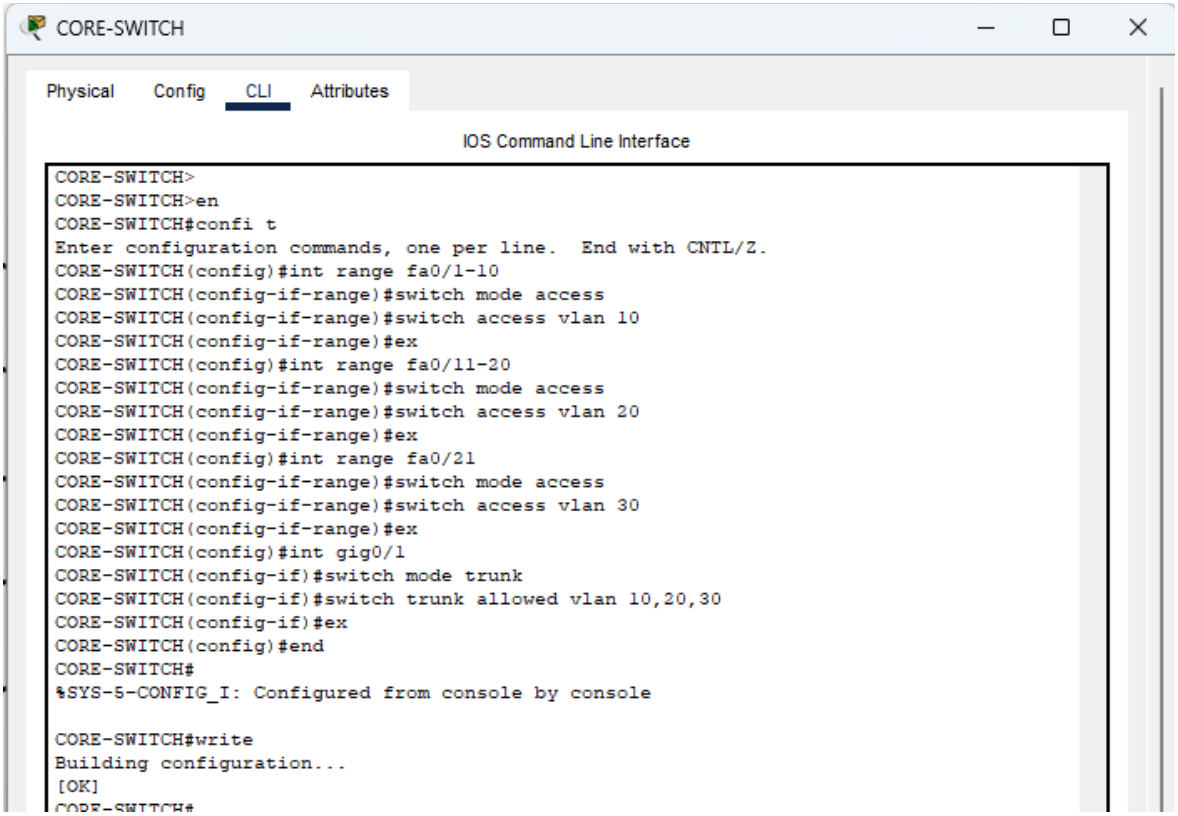
The following screenshots are selected from the supplied project folder. Duplicate screenshots were omitted to keep the report concise and readable.



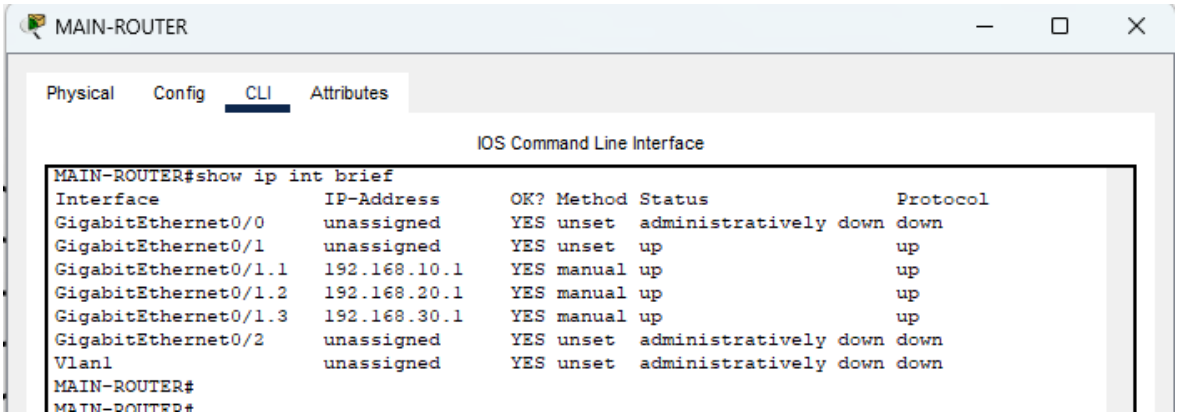
Appendix A: Complete logical network topology.



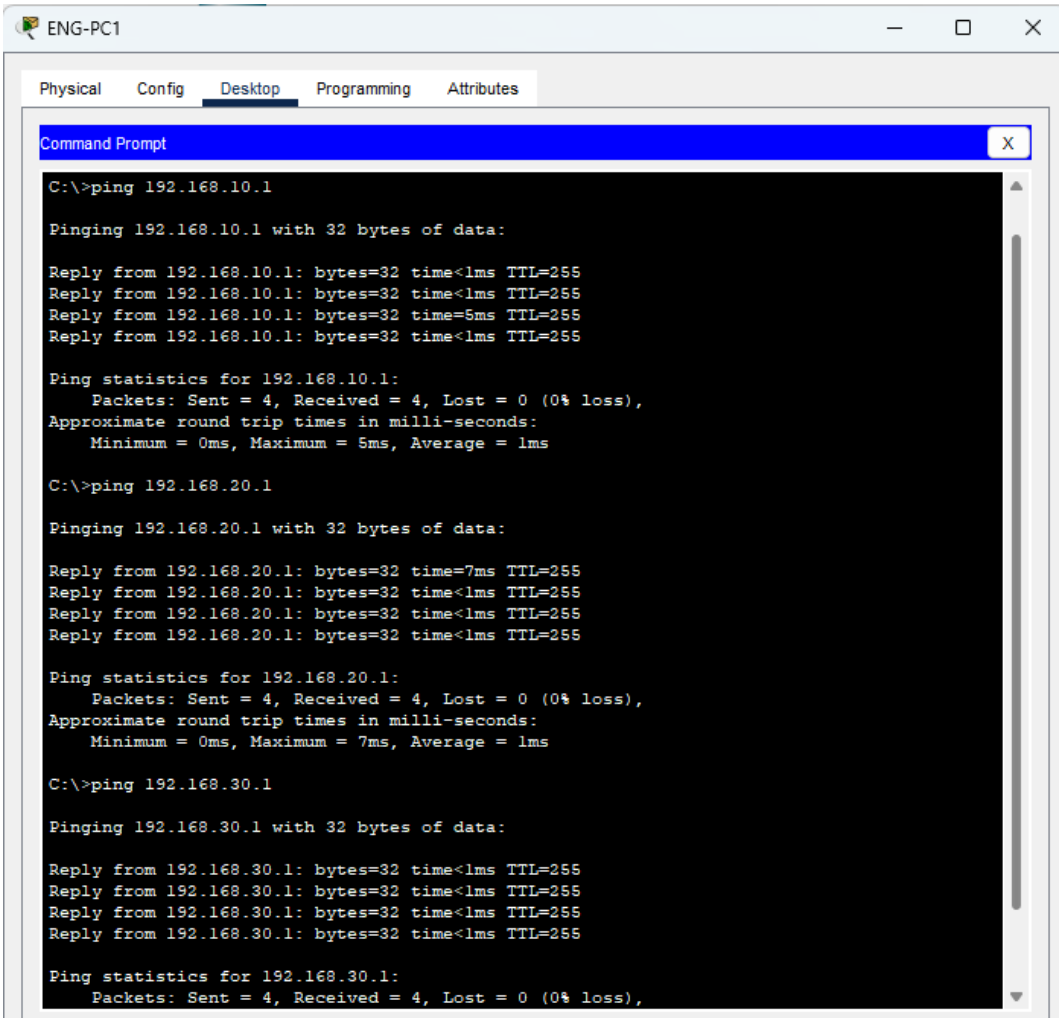
Appendix B: VLAN creation and status.



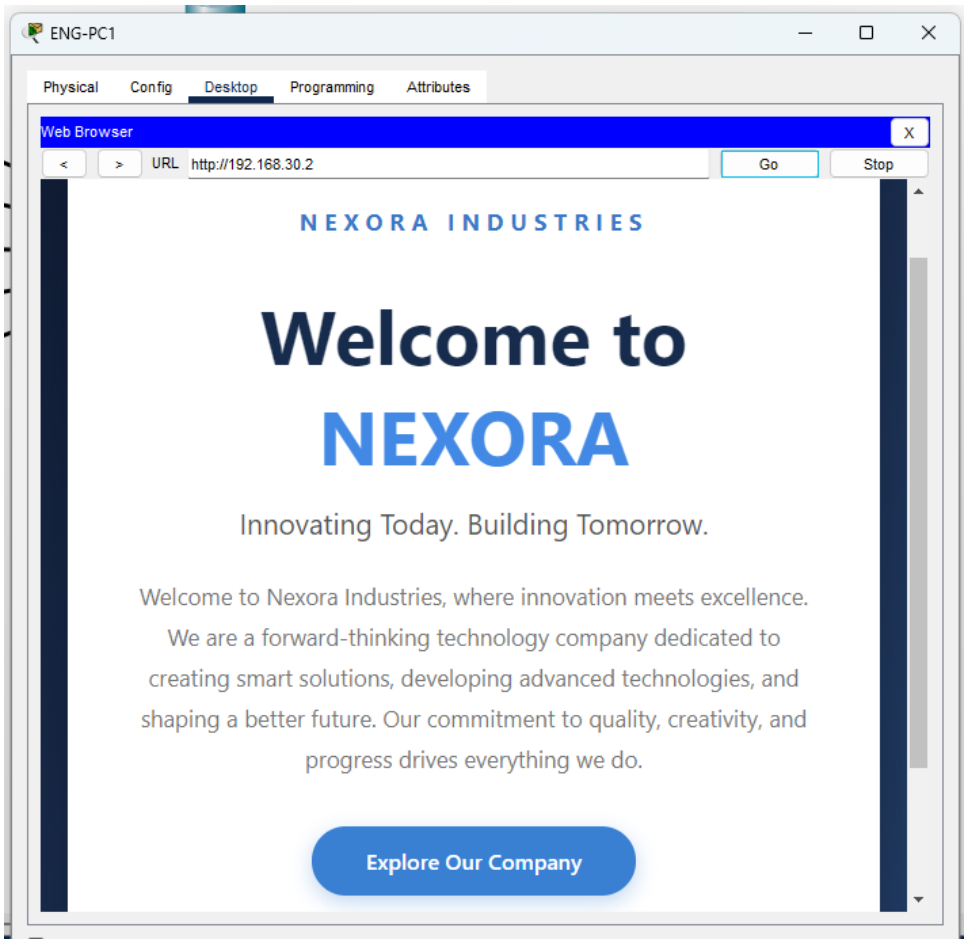
Appendix C: Access port and trunk setup.



Appendix D: Router interface and subinterface status.



Appendix E: PC-side connectivity / command prompt evidence.



Appendix F: Shared server web page / service evidence from the simulation.